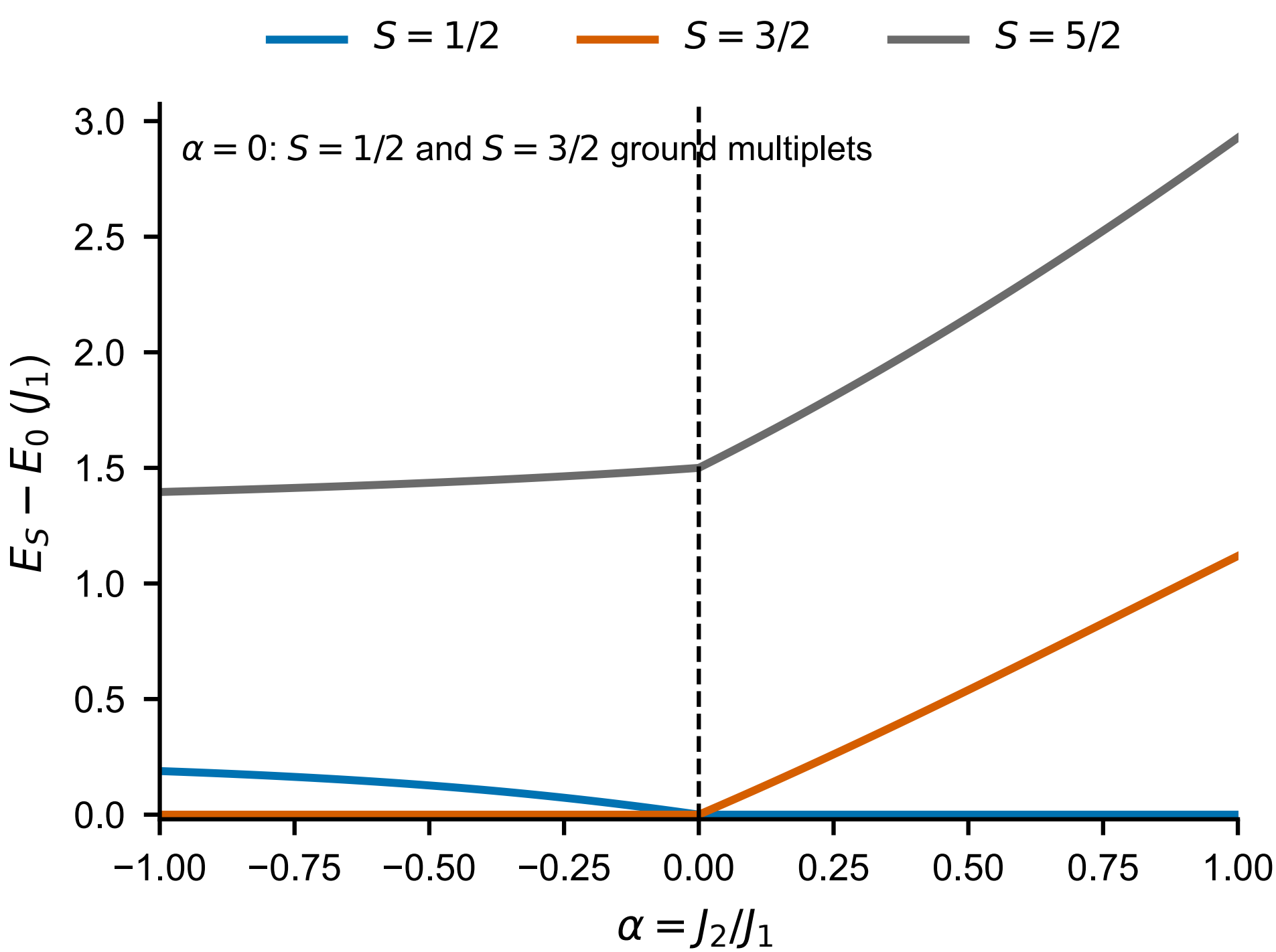
$S = 1/2$
$S = 3/2$
$S = 5/2$
$\alpha = 0$: $S = 1/2$ and $S = 3/2$ ground multiplets
$E_S - E_0\ (J_1)$
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
$\alpha = J_2/J_1$